

Addressing Mechanisms and Heterogeneity in Change with Usable Tools

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<https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk>

Training and mentorship



The Ateneo de Manila University



Liane Peña-Alampay



University of Macau



Shu Fai Cheung



PennState.

The Pennsylvania State University



Sy-Miin Chow



Michael A.
Russell

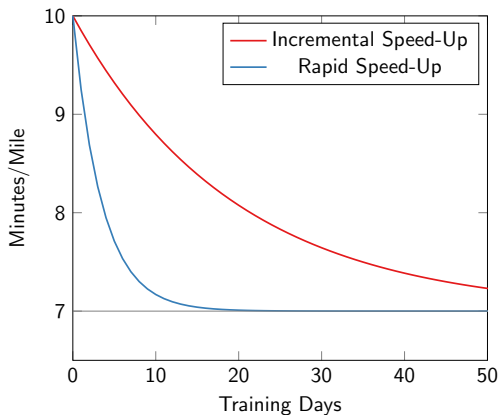
My work integrates **developmental science**, **quantitative methods**, and **dynamic modeling** to study mechanisms and heterogeneity in change across development, family life, and health—with **rigorous, usable methods**.

An intuition for inertia



Inertia: how much the past carries forward into the present.

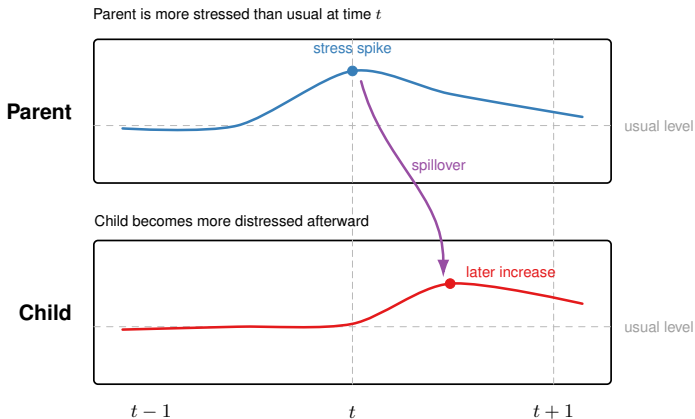
Inertia changes trajectories



Inertia: The same starting point can produce very different trajectories when the **past carries forward** with different strength.

Coupling and spillover

Escalation / Spillover



Coupling: change in one process predicts later change in another.



Inferences and Effect Sizes for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Ivan Jacob Agaloos Pesigan¹, Michael A. Russell^{1, 2}, and Sy-Miin Chow³

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²Department of Biobehavioral Health, The Pennsylvania State University

³Department of Human Development and Family Studies, The Pennsylvania State University

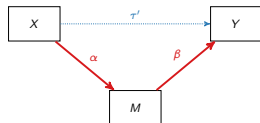
Question: How do mediation effects unfold across time when the process is modeled continuously rather than at one arbitrary lag?

Why mediation has been a recurring part of my work

- ▶ **Substantive** (e.g.; Bernardo, Wang, Pesigan, & Yeung, 2017; Pesigan, Luyckx, & Alampay, 2014; Zhang, Ku, Wu, Yu, & Pesigan, 2020)
- ▶ **Methodological** (e.g.; Cheung & Pesigan, 2023b; Cheung, Pesigan, & Vong, 2023; Pesigan & Cheung, 2020, 2024; Pesigan, Sun, & Cheung, 2023).
 - ▶ <https://CRAN.R-project.org/package=semccci>
 - ▶ <https://CRAN.R-project.org/package=semblci>
 - ▶ <https://CRAN.R-project.org/package=betaDelta>
 - ▶ <https://CRAN.R-project.org/package=betaDSandwich>

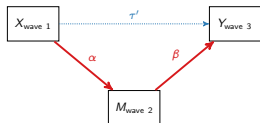
Study 1 extends that line of work into continuous time by adding **standardized effect sizes**, principled **uncertainty quantification**, and an open-source tool in cTMed.

Why this problem matters



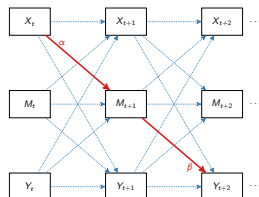
Cross-sectional

One snapshot



Three-wave

Coarse temporal ordering



Discrete-time vector
autoregressive model

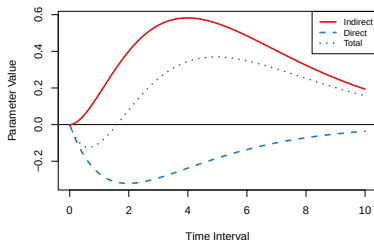
Repeated carryover across lags

- ▶ Mediation is fundamentally a **temporal process**.
- ▶ But in discrete time, the size of the effect depends on the **chosen measurement interval**.

The **continuous-time** shift is to model the process once, then trace how direct, indirect, and total **effects unfold over any time interval**, including irregularly spaced designs.

Novel contributions

Total, Direct, and Indirect Effects

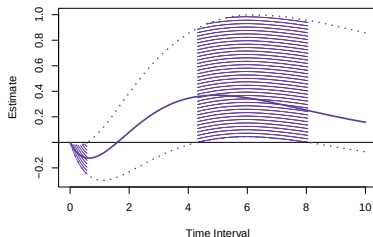


Effect size measures

Standardized total, direct, and indirect effects

- ▶ <https://CRAN.R-project.org/package=cTMed>
- ▶ <https://CRAN.R-project.org/package=simStateSpace>
- ▶ <https://CRAN.R-project.org/package=bootStateSpace>

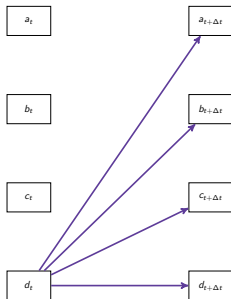
95% CI for the Total Effect (Monte Carlo Method)



Uncertainty quantification

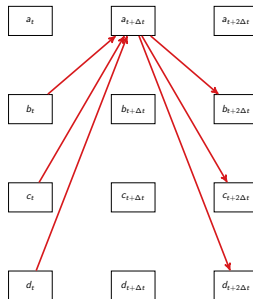
Regions of significance (delta, Monte Carlo, and parametric bootstrap)

One direction this line of work opens up



Total effect centrality

Which variable matters most for shaping the rest of the system over time?



Indirect effect centrality

Which variable matters most as the pathway through which effects spread over time?

This framework can also be extended from single mediated pathways to **system-level summaries** of how effects spread through dynamic networks over time.

Ongoing work on person-specific affect networks: Pesigan, Russell, Chow, Fisher, & Barbara L. Fredrickson, invited submission to an *Emotion* special issue.



Common and Unique Latent Transition Analysis (CULTA) as a Way to Examine the Trait–State Dynamics of Alcohol Intoxication

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Question: Can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?

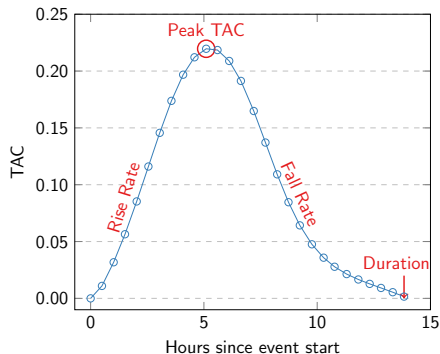
Why this substantive problem demanded a new model

Why Study 2 is exciting

- ▶ **Substantive grounding:** Alcohol and addiction have been a consistent part of my applied work (e.g. Yu, Pesigan, Zhang, & Wu, 2019; Yu, Wu, & Pesigan, 2015; Zamboanga, Iwamoto, Pesigan, & Tomaso, 2015; Zamboanga, Pesigan, Tomaso, Schwartz, et al., 2015; Zhang, Ku, Wu, Yu, & Pesigan, 2020; Zhang, Pesigan, Kahler, Yip, et al., 2019).
- ▶ **Substantive question:** How do risky drinking patterns persist or change across days?
- ▶ **Methodological challenge:** Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.

Substantive grounding in addiction research,
together with the quantitative expertise to build
new models when the science demands them.

From TAC curves to dynamic questions

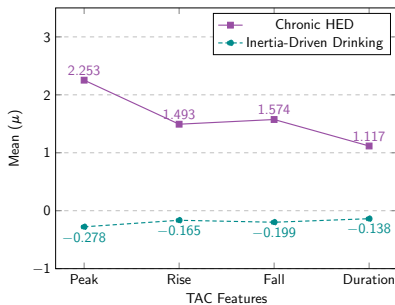


Simple summaries of the TAC features miss three things:

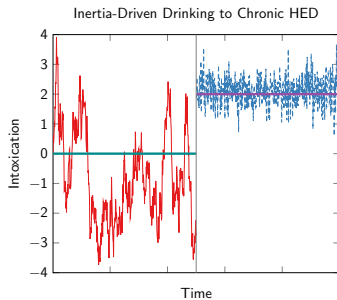
- ▶ what is **shared** across features,
- ▶ what is **feature-specific**,
- ▶ and whether intoxication shows **inertia** across days.

The question is not just **how intoxicated was this day?** but also **what kind of drinking pattern does it reflect, and how stable is that pattern over time?**

Profiles differed in mean intoxication and inertia



Difference in mean levels



Difference in temporal persistence

- ▶ Chronic HED: Higher intoxication mean and Lower inertia
- ▶ Inertia-driven drinking: Lower intoxication mean and Higher inertia

The key point is that these profiles are **not** defined by means alone. They also differ in **how** strongly intoxication carries forward across days.

AUD risk predicts getting stuck in chronic HED

Start in chronic HED

No risk $\rightarrow$ Risk $\rightarrow$ Likely
AUD

$$0.03 \rightarrow 0.07 \rightarrow 0.15$$

Stay in chronic HED

No risk → Risk → Likely
AUD

$$0.21 \rightarrow 0.30 \rightarrow 0.40$$

Shift into chronic HED

No risk $\rightarrow$ Risk $\rightarrow$ Likely
AUD

$$0.03 \rightarrow 0.06 \rightarrow 0.10$$

CULTA does not just classify drinking days; it helps identify **who is likely to get stuck in riskier dynamic patterns.**

Study 3

Under review at *Multivariate Behavioral Research*

From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

Ivan Jacob Agaloos Pesigan¹, Michael A. Russell^{1,2}, and Sy-Miin Chow³

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²Department of Biobehavioral Health, The Pennsylvania State University

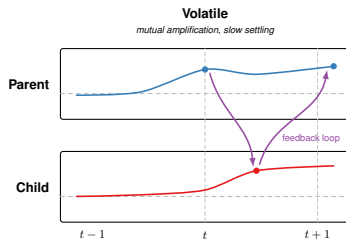
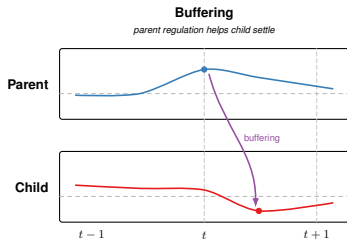
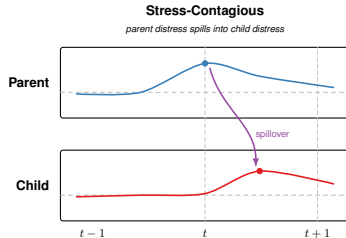
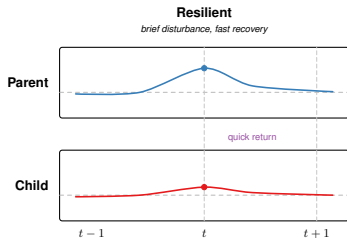
³Department of Human Development and Family Studies, The Pennsylvania State University

Question: How can we draw **population-level** conclusions from **person-specific dynamics** without averaging away the very individual differences we care about?

Dyadic phenotypes

Dynamic Phenotype Profiles of Dyadic Functioning

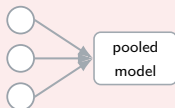
Illustrative parent-child dynamic signatures



The trade-off MetaVAR is designed to solve

Population inference

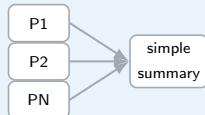
One-step hierarchical models



- + strong for average dynamics
- partial pooling can shrink person-specific dynamics

Idiographic fidelity

Naive two-step workflow



- + keeps person-specific fits
- naive summaries ignore how uncertain those fits are

MetaVAR: estimate each person separately, then synthesize them while carrying **first-stage uncertainty** forward

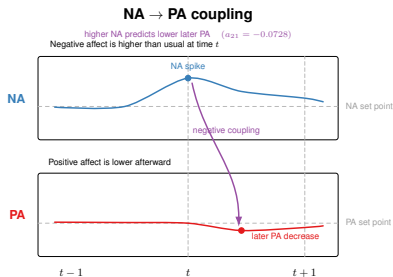
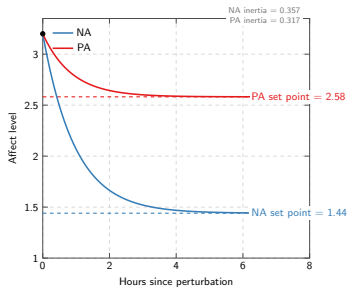
Simulation results: MetaVAR is strongest for heterogeneity

Coverage heatmap by parameter, design cell, and method

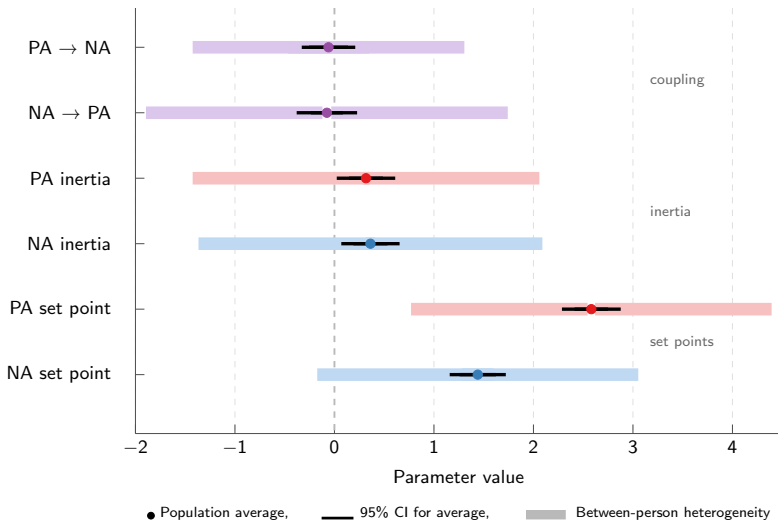
Negative values indicate undercoverage; zero indicates nominal coverage.



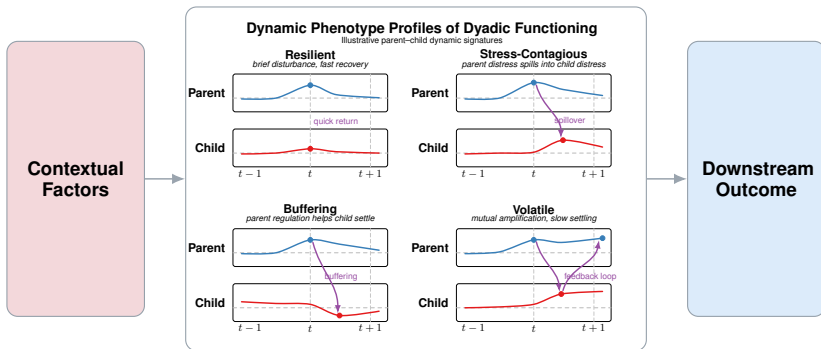
Empirical illustration: Average affect dynamics



Empirical illustration: Heterogeneity in affect dynamics

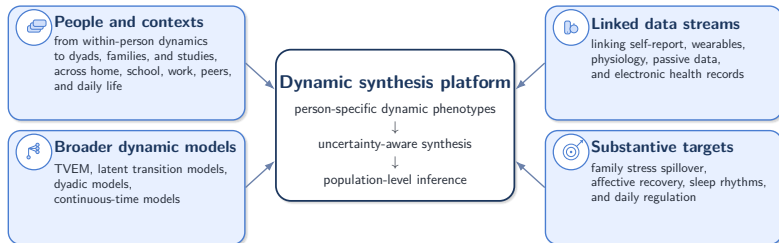


Examining heterogeneity and downstream effects

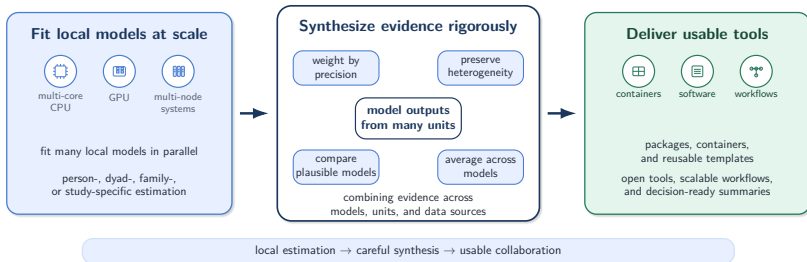


- ▶ <https://CRAN.R-project.org/package=metaDyn>
 - ▶ `meta <- MetaVARmX(estimates, x = contextual, z = downstream)`

Future directions: Toward dynamic phenotypes in HDFS



Future directions: Making dynamic synthesis usable



Matri meae, VAP.

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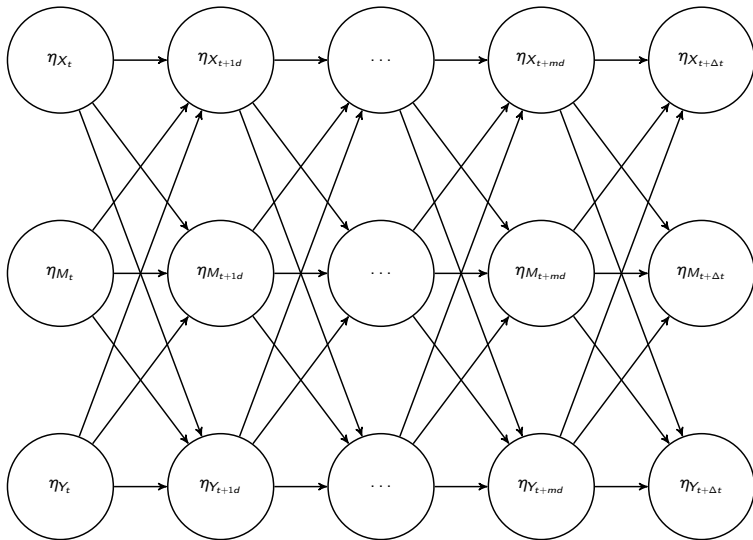


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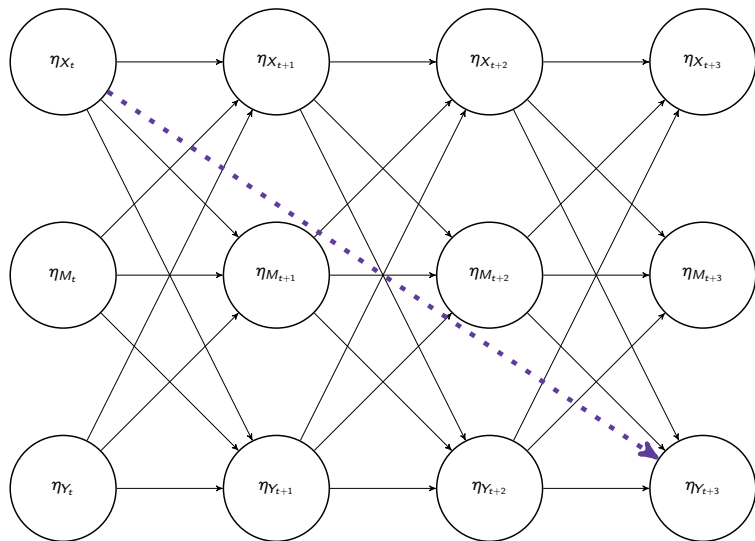
<https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk>

Extra slides

Continuous-Time Vector Autoregressive Model (CT-VAR: Drift Matrix Φ)

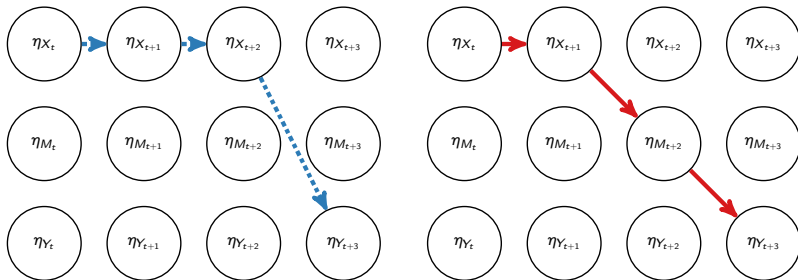


Total Effect Decomposition (Deboeck & Preacher, 2015; Pesigan, Russell, & Chow, 2025b; Ryan & Hamaker, 2021)



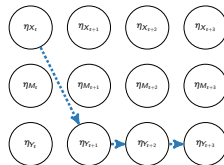
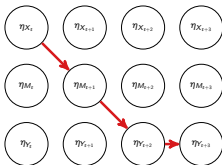
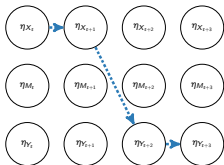
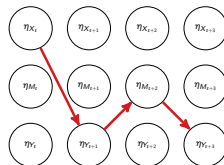
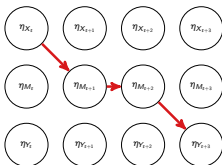
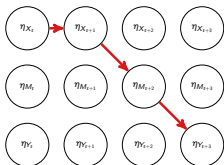
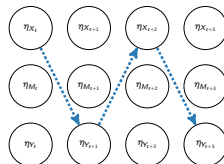
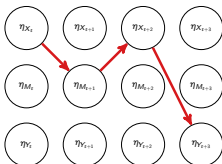
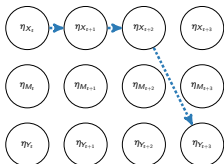
Total Effect Decomposition

Blue = direct effect ($\beta_{11}\beta_{11}\beta_{31}$). Red = indirect effect ($\beta_{11}\beta_{21}\beta_{32}$).



Total Effect Decomposition

Blue = direct effects. Red = indirect effects.



Total Effect Decomposition

Total, Direct, and Indirect Effects:

$$\text{Total} = \exp(\Delta t \Phi) = \beta_{\Delta t} \quad (1)$$

$$\text{Direct} = \exp(\Delta t \mathbf{D}_m \Phi \mathbf{D}_m) \quad (2)$$

$$\text{Indirect} = \text{Total} - \text{Direct}. \quad (3)$$

Standardized Effects: Effect Size Measures

$$\text{Total}_{i,j}^* = \text{Total}_{i,j} \left(\frac{\sigma_{x_j}}{\sigma_{y_i}} \right), \quad (4)$$

$$\text{Direct}_{i,j}^* = \text{Direct}_{i,j} \left(\frac{\sigma_{x_j}}{\sigma_{y_i}} \right), \quad \text{and} \quad (5)$$

$$\text{Indirect}_{i,j}^* = \text{Total}_{i,j}^* - \text{Direct}_{i,j}^* \quad (6)$$

CT-VAR (Chow et al., 2023; Deboeck & Preacher, 2015)

Measurement component:

$$\mathbf{y}_{i,t_m} = \boldsymbol{\nu} + \mathbf{\Lambda}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \boldsymbol{\varepsilon}_{i,t_m} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}) \quad (7)$$

Dynamic component:

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Sigma}^{\frac{1}{2}}d\mathbf{W}_{i,t} \quad (8)$$

Three-variable CT-VAR mediation model:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix} \quad (9)$$

Integral form:

$$\boldsymbol{\eta}_{i,t_{m_i}} = \boldsymbol{\alpha}_{\Delta t_{m_i}} + \boldsymbol{\beta}_{\Delta t_{m_i}}\boldsymbol{\eta}_{i,t_{m_i}-1} + \boldsymbol{\zeta}_{i,t_{m_i}}, \boldsymbol{\zeta}_{i,t_{m_i}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}) \quad (10)$$

$$\boldsymbol{\beta}_{\Delta t_{m_i}} = \exp(\Delta t \boldsymbol{\Phi}) \quad (11)$$

Delta Method Confidence Intervals

We define the delta method following the multivariate central limit theorem. Let $\boldsymbol{\theta}$ be $\text{vec}(\boldsymbol{\Phi})$, that is, the elements of the drift matrix $\boldsymbol{\Phi}$ in vector form sorted column-wise. Let $\hat{\boldsymbol{\theta}}$ be $\text{vec}(\hat{\boldsymbol{\Phi}})$, that is, the estimates of the elements of the drift matrix. The uncertainty around any functions of the estimates for $\hat{\boldsymbol{\theta}}$ is assumed to be characterized as:

$$\sqrt{n} \left(\mathbf{g}(\hat{\boldsymbol{\theta}}) - \mathbf{g}(\boldsymbol{\theta}) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J}\boldsymbol{\Gamma}\mathbf{J}') \quad (12)$$

where $\mathbf{g}$ is a vector of functions, $\mathbf{J}$ is the matrix of first-order derivatives of the functions in $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. The approximate distribution of $\mathbf{g}(\hat{\boldsymbol{\theta}})$ is given by

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) \approx \mathcal{N}(\mathbf{g}(\boldsymbol{\theta}), n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}') . \quad (13)$$

When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}(\hat{\boldsymbol{\theta}})$ for $n^{-1}\boldsymbol{\Gamma}$.

Monte Carlo (MC) Method Confidence Intervals

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters

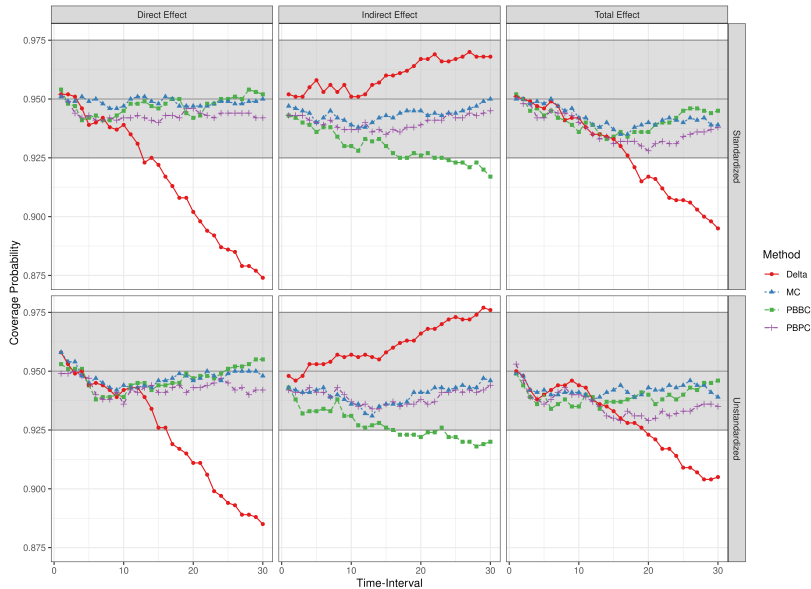
$$\hat{\theta} \sim \mathcal{N}\left(\theta, \mathbb{V}\left(\hat{\theta}\right)\right). \quad (14)$$

Using this distributional assumption, a sampling distribution of $\hat{\theta}$ which we refer to as $\hat{\theta}^*$ can be generated by replacing the population parameters with sample estimates, that is,

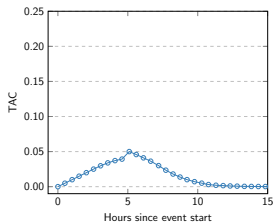
$$\hat{\theta}^* \sim \mathcal{N}\left(\hat{\theta}, \hat{\mathbb{V}}\left(\hat{\theta}\right)\right). \quad (15)$$

A sampling distribution of $\mathbf{g}\left(\hat{\theta}\right)$, which we refer to as $\mathbf{g}\left(\hat{\theta}^*\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the SEs. In resampling approaches, it is customary to use the percentiles of the generated distribution as CIs instead of the SEs.

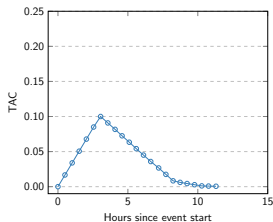
Study 1: Illustration Results



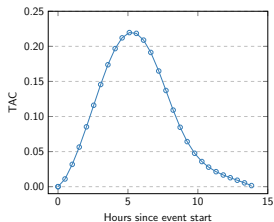
TAC curves for a specific individual



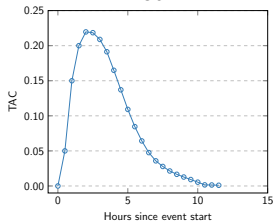
Wed



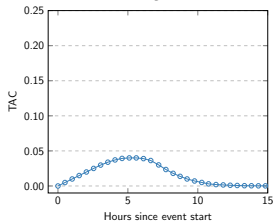
Thu



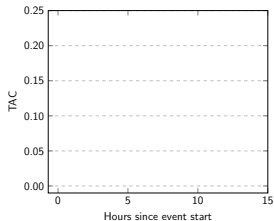
Fri



Sat

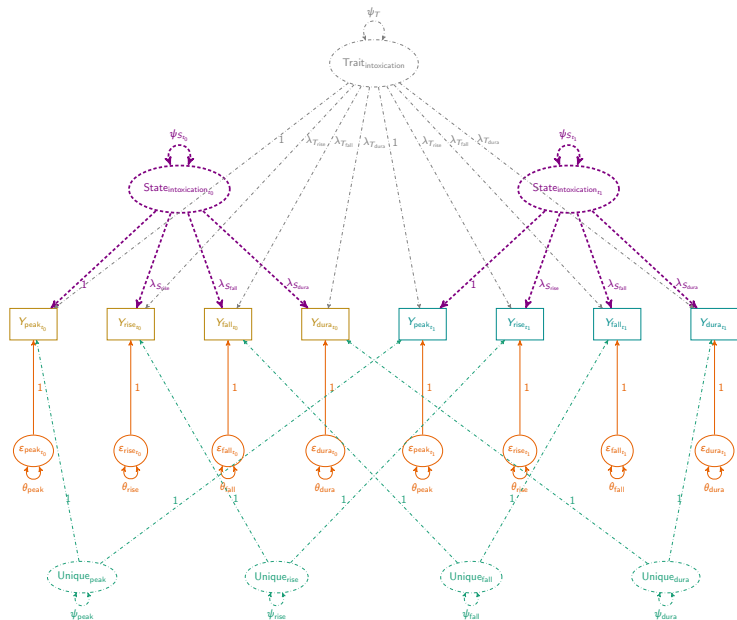


Sun

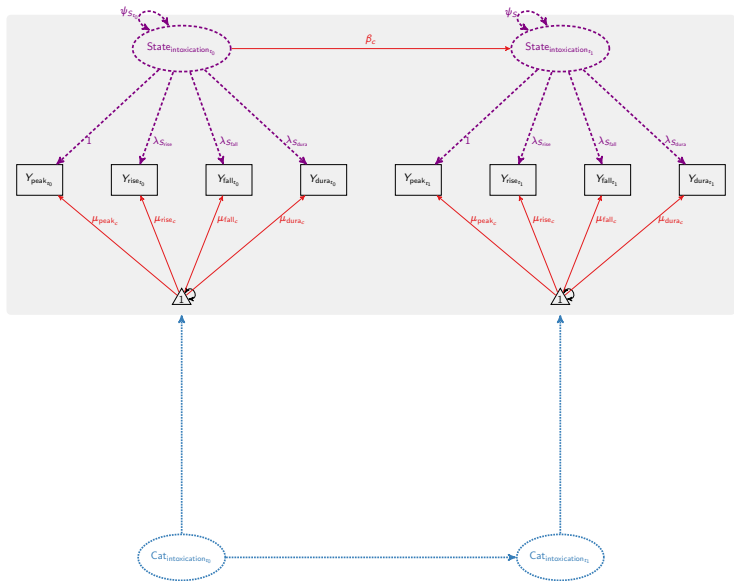


Mon

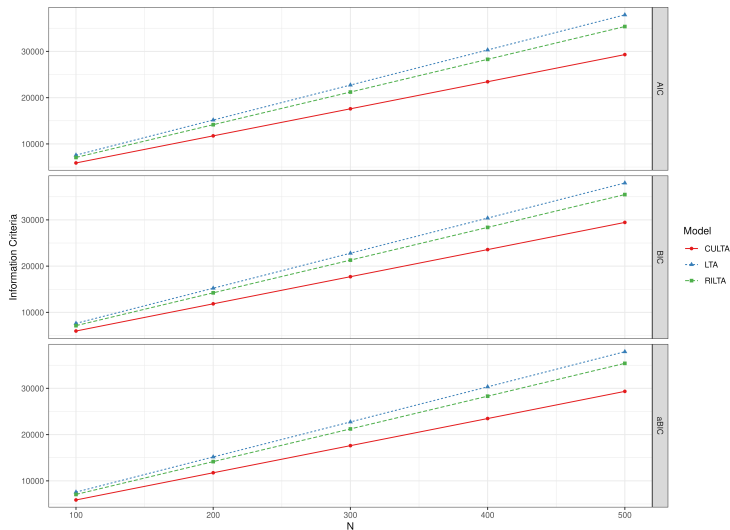
CUTS component of the CULTA model



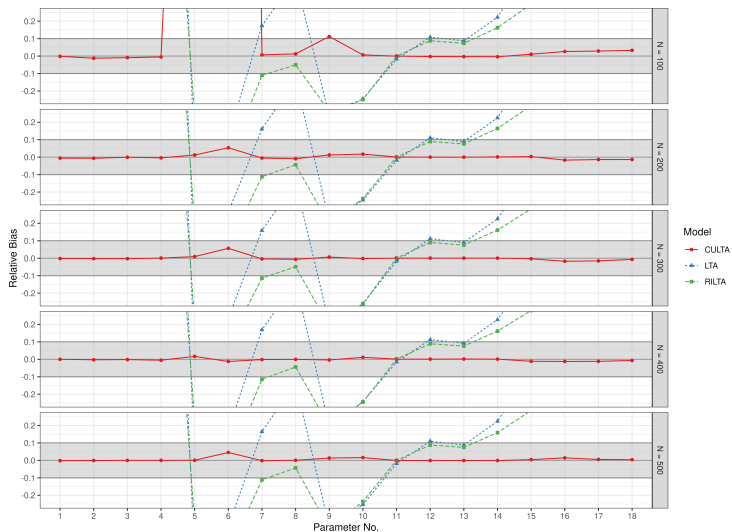
LTA component of the CULTA model



CULTA simulation: Information criteria



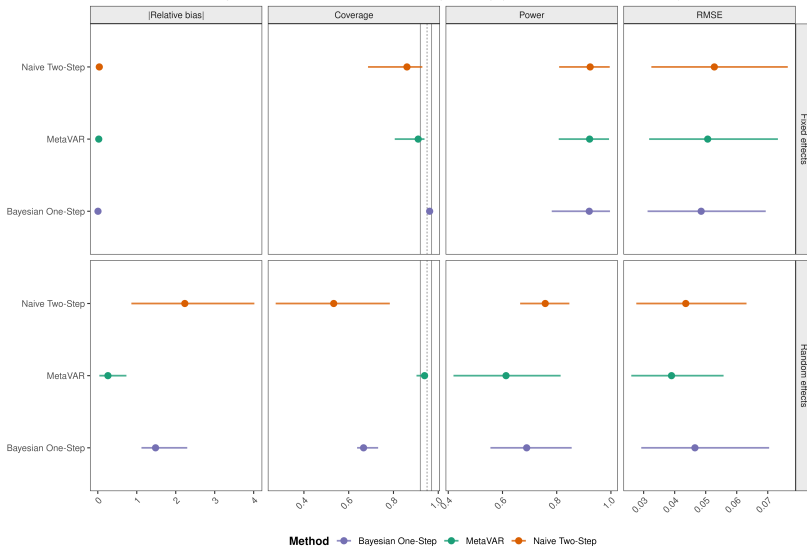
CULTA simulation: Relative bias



Simulation results: MetaVAR is strongest for heterogeneity

Overall method performance across simulation design cells

Points are means across design cells; horizontal lines show the min-max range. Coverage panels include nominal .95 and Bradley's liberal .92-.97 band.

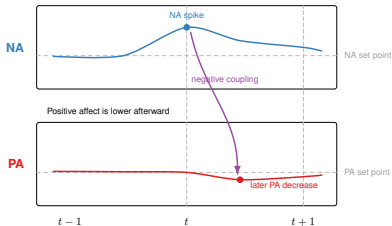


Empirical illustration: Coupling

NA → PA coupling

higher NA predicts lower later PA ($a_{21} = -0.0728$)

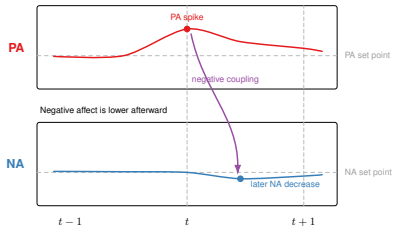
Negative affect is higher than usual at time t



PA → NA coupling

higher PA predicts lower later NA ($a_{12} = -0.0593$)

Positive affect is higher than usual at time t



References

References I



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